

Product Line Profitability & Margin Performance Analysis

Nassau Candy Distributor - Research Paper (EDA, Insights and Recommendations)

Data period: January 2024 - December 2025 | Live dashboard: <https://candy-margin-analysis.streamlit.app/>
| Code: https://github.com/radhikatech006-ui/Nassau_candy_dashboard

Abstract. Sales volume alone does not show which products make a distributor money. This paper analyses 10,194 order lines (8,549 orders, 5,044 customers) of Nassau Candy Distributor, covering 15 products in three divisions, to show which product lines deliver margin, which only deliver revenue, and where margin risk sits. The business earned **\$141,784** in sales and **\$93,443** in gross profit, a gross margin of **65.9%**. Profit is extremely concentrated: five Wonka Bar products generate **95.1%** of gross profit, and the top three alone generate 59.3%. One product, Kazookles, earns a margin of only **7.7%** and is the clearest case for repricing, cost renegotiation or a discontinuation review. Because every product is sold at a fixed price and cost, margin volatility over time is zero: margin risk here is structural (which product is sold), not time-driven. The analysis is delivered as a reproducible notebook and an interactive Streamlit dashboard.

1. Introduction

For a distributor, a product that sells in high volume is not necessarily a product that earns money. Some products consume a disproportionate share of manufacturing or operating cost, and some look successful while quietly pulling down the overall margin. Nassau Candy Distributor currently lacks a clear view of which product lines deliver the highest gross margin, whether high-sales products are actually profitable, how profitability differs across divisions, and which products represent margin risk. Without that view, pricing, promotion and portfolio decisions stay reactive and intuition-based.

This study answers four questions: (1) which products drive profit, (2) which divisions underperform financially, (3) how concentrated profit is and how dependent the business is on a few products, and (4) which products need repricing, cost renegotiation or a discontinuation review.

2. Data and Cleaning

The dataset contains 10,194 order-line records and 18 fields (order and shipping dates, customer geography, division, product, sales, units, cost and gross profit) from 2 January 2024 to 31 December 2025. Each row is one product line within an order. The following validation steps were applied before analysis:

- **Completeness and uniqueness:** no missing values, no duplicate rows, and Row ID is unique.

- **Validity of values:** no zero or negative sales, units, cost or gross profit, so no records needed to be removed.
- **Profit integrity:** Gross Profit equals Sales minus Cost for every row once rounded to two decimals (3,809 exact-match differences were floating-point noise only).
- **Labels:** category fields were checked for stray spaces and inconsistent casing, and Product ID to Product Name consistency was checked.
- **Outliers:** an IQR check flagged some large orders (for example 245 on Sales). These are legitimate larger purchases, so they were documented and kept.
- **Dates:** Order Date is stored day-first (DD-MM-YYYY) and was parsed accordingly. Ship Date was found to be unreliable: its values fall between 2026 and 2030, about 1,321 days after the order on average, which is implausible. Ship Date was therefore excluded, and all time analysis uses Order Date.

Division labels come from the dataset. Fizzy Lifting Drinks appears under Sugar in the data, although the project brief lists it as Other; the dataset's labelling is used throughout. Product-to-factory assignments follow the mapping supplied in the project brief.

3. Methodology

All metrics are computed from order lines and aggregated by product, division, factory, region, state and year.

KPI / method	Definition
Gross Margin (%)	Gross Profit / Sales x 100 (aggregated as total profit / total sales, not an average of row margins)
Profit per Unit	Gross Profit / Units
Revenue / Profit Contribution	Product sales (or profit) / total sales (or profit)
Margin Volatility	Standard deviation of monthly gross margin (percentage points) for each product
Product classification	Median split on total sales and gross margin: High/Low Sales x High/Low Margin
Pareto analysis	Products ranked by revenue and by profit; number needed to reach 80% cumulative share
Dependency indicators	Share of profit from top 1, top 3 and top 5 products; Herfindahl-Hirschman Index (HHI = sum of squared profit shares x 10,000). Risk is HIGH if top-3 share is at least 50%, top-5 share at least 80%, or HHI at least 2,500
Margin risk flags	Margin below 20%: Reprice (plus Discontinuation review if under 2% of revenue). Cost-to-sales more than 10 points above the division average: Cost renegotiation

The risk thresholds (20%, 10 points, 2%, and the dependency cut-offs) are analytical judgements chosen for this study, not industry standards; they are adjustable constants in the code.

4. Results

4.1 Product-level profitability

The company earned \$141,784 in sales on \$48,341 of cost, leaving \$93,443 of gross profit on 38,654 units (an overall margin of 65.9% and \$2.42 profit per unit). Table 1 ranks all 15 products by gross profit.

Table 1. Product profitability, ranked by gross profit (WB = Wonka Bar).

Product	Division	Sales (\$)	Gross profit (\$)	Margin	Profit / unit	Share of profit
WB Scrumdiddlyumptious	Chocolate	27,875	19,358	69.4%	\$2.50	20.7%
WB Triple Dazzle Caramel	Chocolate	28,485	18,610	65.3%	\$2.45	19.9%
WB Milk Chocolate	Chocolate	26,868	17,443	64.9%	\$2.11	18.7%
WB Nutty Crunch Surprise	Chocolate	23,575	16,820	71.3%	\$2.49	18.0%
WB Fudge Mallows	Chocolate	24,890	16,594	66.7%	\$2.40	17.8%
Lickable Wallpaper	Other	7,860	3,930	50.0%	\$10.00	4.2%
Wonka Gum	Other	598	311	52.0%	\$0.65	0.3%
Everlasting Gobstopper	Sugar	130	104	80.0%	\$8.00	0.1%
Kazookles	Other	1,206	93	7.7%	\$0.25	0.1%
Hair Toffee	Sugar	77	60	77.8%	\$3.50	0.1%
Fizzy Lifting Drinks	Sugar	79	47	60.0%	\$2.25	0.05%
Laffy Taffy	Sugar	54	33	62.3%	\$1.24	0.04%
SweetTARTS	Sugar	62	29	46.7%	\$0.70	0.03%
Nerds	Sugar	15	7	46.7%	\$0.70	0.01%
Fun Dip	Sugar	12	5	40.0%	\$0.60	0.01%

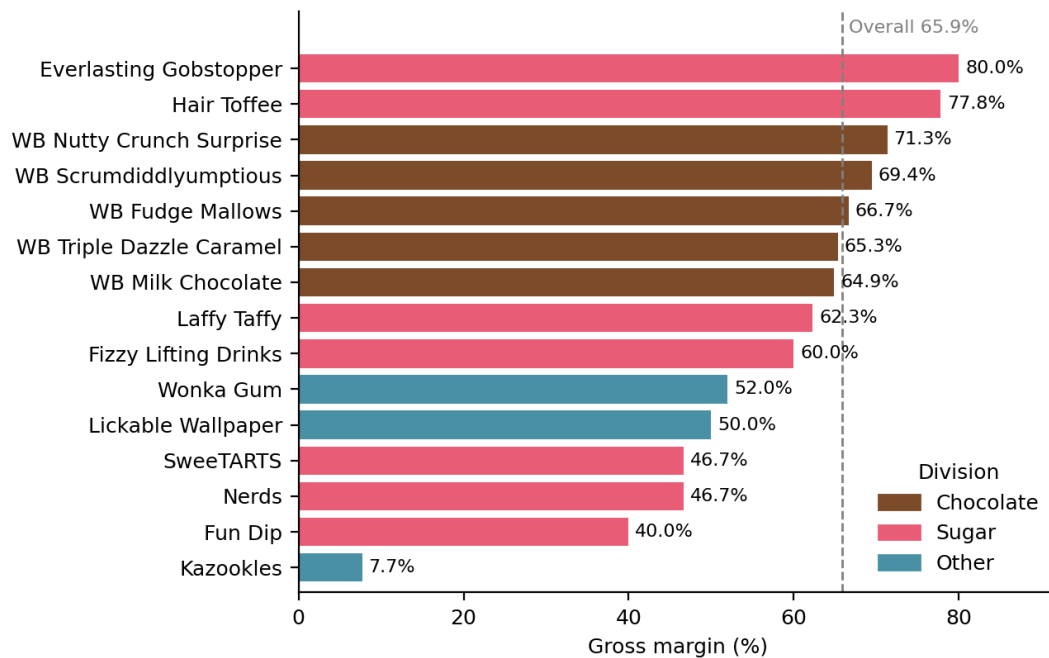


Figure 1. Product margin leaderboard. The five Wonka Bars sit between 65% and 71%; Kazookles is far below every other product.

Three patterns stand out. First, the five Wonka Bars are both the biggest sellers and highly profitable, each earning a margin between 64.9% and 71.3%. Second, high sales do not guarantee high margin: Lickable Wallpaper and Wonka Gum are at or above the median on sales but earn only 50% and 52%, and Kazookles, with \$1,206 of sales (more than 8 of the other 14 products) earns just \$93 of profit. Third,

several Sugar products earn very high margins (Everlasting Gobstopper 80%, Hair Toffee 77.8%) but sell so little that they contribute almost nothing to profit.

Using the median split (median sales \$598, median margin 62.3%), the products fall into four groups:

Group	Products
High sales / high margin	All five Wonka Bars
High sales / low margin	Lickable Wallpaper (50.0%), Wonka Gum (52.0%), Kazookles (7.7%)
Low sales / high margin	Everlasting Gobstopper, Hair Toffee, Laffy Taffy
Low sales / low margin	Fizzy Lifting Drinks, SweetARTS, Nerds, Fun Dip

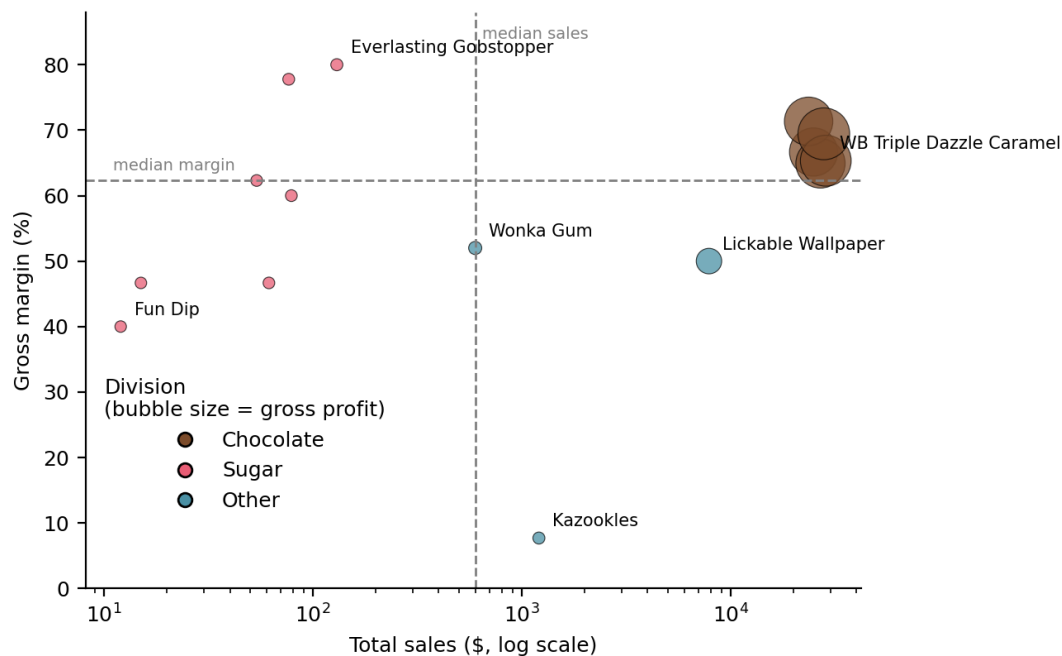


Figure 2. Product profitability matrix (log sales scale). Dashed lines show the median sales and median margin.

Profit per unit tells a different story from margin. Lickable Wallpaper earns \$10.00 per unit, the highest of any product and four times a Wonka Bar, even though its margin is only 50%. It is the only non-chocolate product with a material profit contribution (4.2%).

4.2 Division-level performance

Division	Sales (\$)	Gross profit (\$)	Revenue share	Profit share	Margin (weighted)	Avg. product margin
Chocolate	131,693	88,825	92.9%	95.1%	67.4%	67.5%
Other	9,663	4,333	6.8%	4.6%	44.8%	36.6%
Sugar	427	285	0.3%	0.3%	66.6%	59.1%

Table 2. Division performance (weighted margin = total profit / total sales; average product margin is the simple mean of product margins).

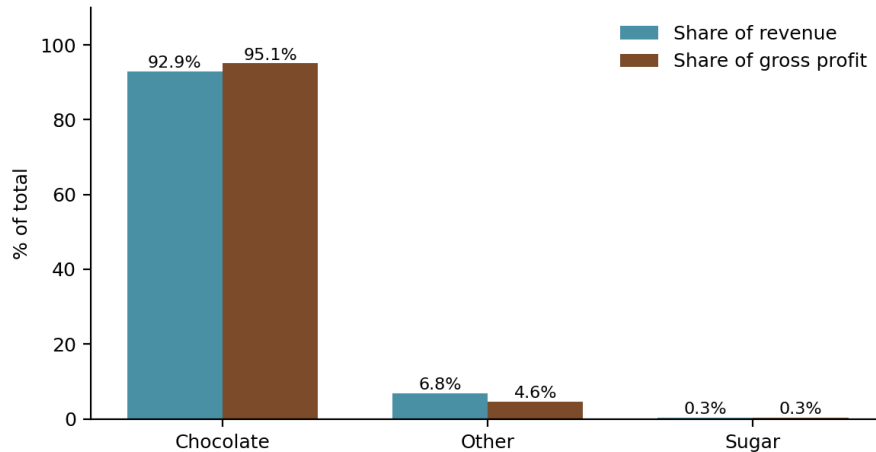


Figure 3. Share of revenue versus share of gross profit by division.

Chocolate converts revenue into profit more efficiently than the business average: it generates 92.9% of revenue but 95.1% of profit. Other does the opposite, producing 6.8% of revenue but only 4.6% of profit, a structural margin problem caused almost entirely by Kazookles (without it, the other two Other products earn about 50-52%). Sugar's weighted margin of 66.6% looks strong, but the division sold only \$427 in two years across 137 units, so the figure is not economically meaningful; its simple average product margin of 59.1% is a better description of the range (40% to 80%).

4.3 Profit concentration (Pareto analysis)

Five of 15 products (33% of the line) are needed to reach 80% of revenue, and the same five are needed to reach 80% of profit. All five are Wonka Bars. Together they produce 92.9% of revenue and 95.1% of gross profit, leaving 10 products to share the remaining 4.9% of profit. The top product earns 20.7% of profit, the top three earn 59.3%, and the profit HHI is 1,831. By the rule in Section 3 the dependency risk is rated **HIGH**.

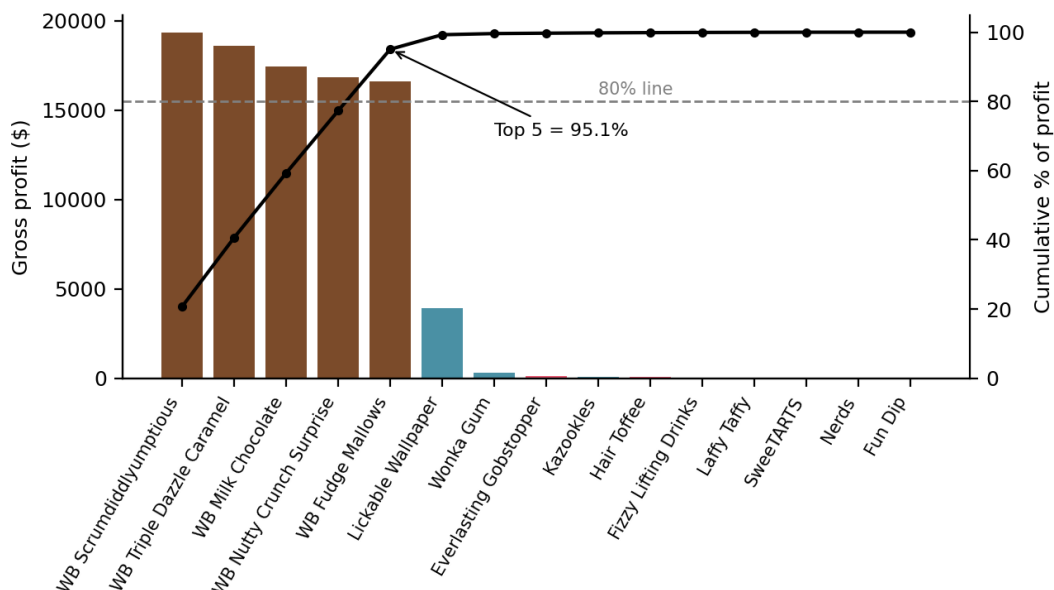


Figure 4. Profit Pareto chart: bars show gross profit by product, the line shows cumulative share of profit.

The dependency is on the Wonka Bar line as a whole rather than on any single product, because the five products are close in size. This means a shock to the chocolate division, such as a supply problem, a price change or a demand shift, would affect almost all profit at once.

4.4 Cost structure diagnostics

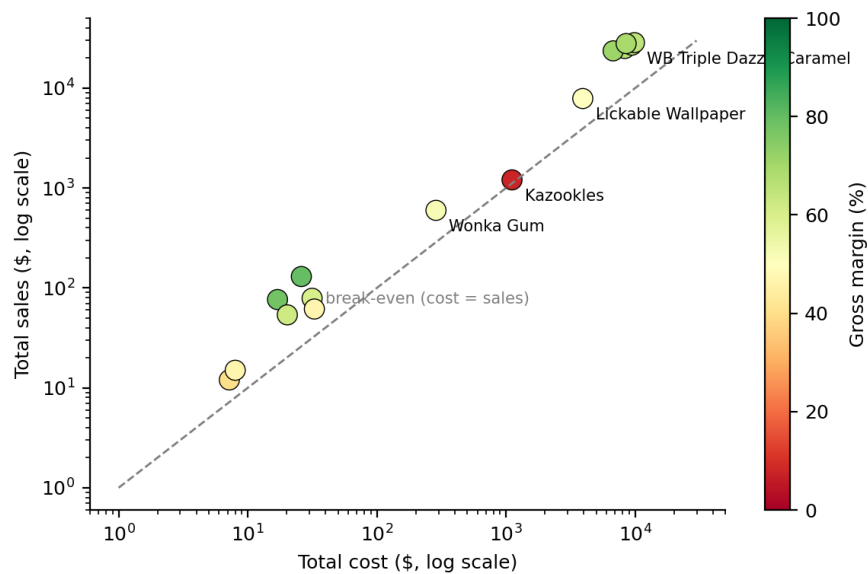


Figure 5. Cost versus sales by product (log scales). Points close to the dashed break-even line earn little margin.

Plotting cost against sales shows the Wonka Bars well above the break-even line with a consistent cost structure, while Kazookles sits almost on it: its cost is 92.3% of sales. Applying the flag rules from Section 3 marks four products:

Product	Margin	Cost-to-sales	Revenue share	Flag
Kazookles	7.7%	92.3%	0.85%	Reprice; Discontinuation review; Cost renegotiation
Fun Dip	40.0%	60.0%	0.01%	Cost renegotiation
Nerds	46.7%	53.3%	0.01%	Cost renegotiation
SweetTARTS	46.7%	53.3%	0.04%	Cost renegotiation

Kazookles sold 371 units at an average price of \$3.25 against a unit cost of \$3.00, a profit of \$0.25 per unit. As an illustration, holding volume and cost constant, a price of about \$3.75 (+15%) would lift its margin to 20%, and reaching the same margin at today's price would need unit cost to fall by about 13% (to \$2.60). These scenarios ignore any change in demand. The money at stake is small (a 20% margin would add roughly \$186 of profit over two years), so the decision is mainly about whether the product deserves shelf and management attention. Fun Dip, Nerds and SweetTARTS cost far more relative to sales than the rest of the Sugar division, but each is below 0.1% of revenue, so cost renegotiation there is a low priority.

4.5 Growth and margin stability over time

Sales grew from \$57,956 in 2024 to \$83,827 in 2025 (+44.6%) and gross profit from \$38,151 to \$55,291 (+44.9%), while units rose 43.1%. Gross margin barely moved (65.83% to 65.96%), so growth did not come at the expense of profitability. Margin volatility, measured as the standard deviation of monthly margin, is exactly zero for all 15 products. Each product is sold at one price and one cost for the whole period, so margins never change. Margin risk is therefore structural: a low-margin product stays low-margin until its price or cost changes.

4.6 Geographic and shipping patterns

Margin does not depend on location. The four regions all earn between 65.5% and 66.4% (Pacific has the most sales at \$46,302, followed by Atlantic at \$41,197, Interior at \$32,038 and Gulf at \$22,247). California is the largest state at \$27,917 in sales (19.7% of the total), followed by New York (\$15,541) and Texas (\$13,416). State margins range from 61.5% to 71.3%, but this spread reflects product mix in small states rather than genuine pricing differences. Ship mode also has no effect on margin (65.7% to 66.1%); Standard Class carries 60.3% of sales. Because every product has a fixed margin, a region or state can only earn a different margin by selling a different mix of products.

4.7 Factory-level performance

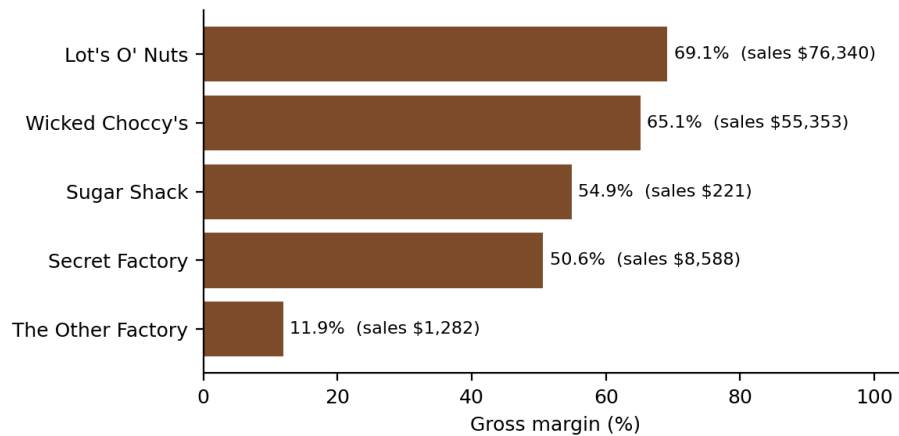


Figure 6. Gross margin by factory (sales in brackets).

Two factories carry the business: Lot's O' Nuts (3 Wonka Bars, \$76,340 in sales, 69.1% margin) and Wicked Choccy's (2 Wonka Bars, \$55,353, 65.1%) together produce 92.9% of sales and 95.1% of profit. Secret Factory earns 50.6% on \$8,588 of sales. The Other Factory earns only 11.9% because it makes Kazookles and Hair Toffee and Kazookles dominates its volume; the weak result reflects one product, not the factory as a whole. This also makes the dependency risk operational: an outage at either chocolate factory would hit most of the company's profit.

5. Key Insights

- **Profit is concentrated in five products.** The Wonka Bars produce 95.1% of gross profit; the other 10 products are almost irrelevant to profit today.
- **Revenue and profit leaders are the same products.** Unlike many distributors, high sales here are not hiding low profit among the leaders. The margin problems are in small products.
- **Kazookles is the one real margin problem.** A 7.7% margin and cost at 92.3% of sales make it the clearest repricing or discontinuation candidate, although its absolute profit effect is small.
- **Margin risk is structural, not time-driven.** With fixed prices and costs, margin volatility is zero; location and shipping mode do not change margin either.
- **Lickable Wallpaper is the main diversification asset.** It earns \$10 per unit and 4.2% of profit, the only non-chocolate product with material contribution.
- **High-margin Sugar products are too small to judge.** Their margins look attractive, but their volumes are tiny, so conclusions about them are not reliable.

6. Recommendations

#	Action	Why
1	Protect the Wonka Bar line: secure supply and capacity at Lot's O' Nuts and Wicked Choccy's, and avoid discounting that erodes the 65-71% margins.	95.1% of profit depends on these five products and two factories.
2	Decide on Kazookles within one quarter: reprice by about 15% (to roughly \$3.75) or cut unit cost by about 13%; if neither is possible, move to a discontinuation review.	7.7% margin; cost is 92.3% of sales.
3	Test growth of Lickable Wallpaper (promotion, wider distribution) and track its profit per unit.	\$10.00 profit per unit, highest of all products.
4	Reduce dependency over time by building one or two further products toward meaningful volume; monitor the dependency indicators monthly in the dashboard.	Dependency risk is HIGH (top-5 share 95.1%, HHI 1,831).
5	Review costs for Fun Dip, Nerds and SweetTARTS only after Kazookles; test whether the high-margin Sugar products can grow before judging them.	Flagged for cost, but each is below 0.1% of revenue.
6	Improve the data: correct Ship Date capture, and record price and cost changes so that margin trends and price tests can be measured.	Ship Date unusable; volatility is zero because prices never change.

7. Dashboard Deliverable

The analysis is available as an interactive Streamlit dashboard at <https://candy-margin-analysis.streamlit.app/> (source code at https://github.com/radhikatech006-ui/Nassau_candy_dashboard). Users can filter by date range, division and product search, and set a minimum-margin threshold, applied at product level. The dashboard contains a product profitability overview (margin leaderboard, contribution charts, profitability matrix), a division performance view (revenue versus profit and the margin distribution by division), cost versus margin diagnostics (cost-sales scatter, cost-to-sales ratios and margin risk flags), profit concentration analysis (revenue and profit Pareto charts with cumulative lines and dependency indicators), and additional geographic and factory sections.

8. Limitations

- The data covers only 15 products and two years, and several products have very few orders, so results for small products are unstable.
- Gross margin is based on manufacturing cost only; shipping, promotion and operating costs are not included, so true profitability is lower.
- Prices and costs are constant for each product, so price sensitivity and margin trends cannot be estimated, and the repricing scenarios assume unchanged demand.
- Ship Date is unusable, so no delivery or logistics analysis was possible.
- Median thresholds, the 20% margin line and the dependency cut-offs are judgement-based; different thresholds could change which products are flagged.

9. Conclusion

Nassau Candy earns a healthy 65.9% gross margin, but it depends on a very narrow base: five Wonka Bars produce 95.1% of gross profit. The sales leaders are also the profit leaders, so the main margin risks are a single weak product (Kazookles) and the concentration itself. Management should protect the chocolate line, fix or exit Kazookles, build a second source of profit (starting with Lickable Wallpaper), and track margin and dependency monthly using the dashboard.